

Public Economics (ECON 131)

Section #10: Social Insurance

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1 Key Concepts

1.1 Insurance Intro

- **Social insurance programs** are government interventions in the provision of insurance against adverse events.
 - ↳ Medicare, Medicaid, Obamacare
 - Eg: health insurance, disability insurance, unemployment insurance retirement insurance (social security)
 - Government provision can address market failures. May also be motivated by desire to redistribute, provide services that are considered a basic right, address individual failures, or reduce administrative costs
- These insurance systems allow for consumption smoothing across states of the world.
 - ↳ for example you pay into unemployment insurance when you are employed and receive money when you are unemployed
- **Insurance premiums:** Money that is paid to an insurer so that an individual will be insured against adverse events.

1.2 Expected Utility Model

the sum of utilities associated w/ all possible outcome (or states), each weighted by the probability of the outcome occurring

- Individuals maximize expected utility (EU) over possible states of the world, where $q = \text{Pr}(\text{adverse event})$:

$$EU = (1-q) \cdot U(\text{consumption w/ no adverse event}) + q \cdot U(\text{consumption w/ an adverse event})$$

- An insurance premium is actuarially fair if it is set equal to the insurer's expected payout.

what are the components of consumption?

- income W (make this from working)
- pay premium p (pay into the system)
- benefit b (what you get from paying into the system)
- cost d $\rightarrow$ the cost of the adverse event
if sick there are medical costs
if unemployed, you lose income

$$EU = (1-q)U(W-p) + qU(W-p-d+b)$$

so, what is the expected profits of insurers?

$$EP = p - qb$$

what you pay the benefit times prob of getting it

$EP = 0$ if competition, then

$$0 = p - qb \Rightarrow p = qb \quad b = \frac{p}{q}$$

actuarially fair insurance

individuals choose the level of premiums p to max
set $b = p/q$, then

$$EU = (1-q)U(W-p) + qU(W-p-d+p/q)$$

$$\frac{\partial EU}{\partial p} = -(1-q)U'(W-p) + q[-1 + 1/q]U'(W-p-d+p/q)$$

$$\Rightarrow U'(W-p) = U'(W-d-p+p/q)$$

$$\Rightarrow W-p = W-d-p+p/q$$

$$\Rightarrow d = p/q \Rightarrow p = dq$$

b/c U is concave and hence U' is strictly decreasing and hence invertible

want premium equal to costs of adverse event times probability of incurring costs

this implies that the person is (fully) perfectly insured:
consumption is the same in both states + equal to $W-d \cdot q$

- Asymmetric information: individuals have private information about their risk factors.

market failure

- **Adverse selection:** individuals with higher risk are more likely to purchase insurance. This makes insurance costlier, and may make the market unravel

- **Moral hazard:** individuals may change their behavior in response to being insured

desirable to reduce consumption in high income states to increase consumption in low income states

$$(b=d)$$

$$\Rightarrow$$

$$EU = (1-q)U(w-p) + qU(w-p) = U(w-p)$$

consumption smoothing!

2 Examples

2.1 Adverse Selection

Do health insurance and car insurance markets both suffer from adverse selection? Are there reasons why the government should intervene in the health insurance market and not in the car market?

2.2 Moral Hazard

Unemployment insurance in the US typically pays the unemployed 50% of their previous wage for about 6 months. During the recession of 2009 it was extended temporarily to a year. Why is unemployment insurance far from 100% replacement? Why did government decided to extend UI during the recession?

3 Practice: Disability Insurance

Consider an economy where there are three types of people who want to purchase disability insurance. Each type has the same health-based risks, such that they each have a 2 percent chance of being incapacitated due to health risks, which are uncorrelated with their risk taking behavior. However, each type of person differs in their hobbies and work in the following manner:

- Low-risk types walk to work and have very low-risk hobbies. Their outside risk of being incapacitated is 3 percent.
- Medium-risk types drive to work and actively play soccer on the weekends. Therefore, their non-health risk of being incapacitated is 8 percent.
- High-risk types bike to work, work as firefighters, and skydive on weekends. Their outside risk of being incapacitated is 58 percent.

Assume that there are equal numbers of each type. Long-term care insurance provides income if they are incapacitated for the rest of their life (there are no additional costs). Suppose individuals have the following utility function over consumption (or income):

$$u(c) = \log(c)$$

Individuals earn \$500 if healthy, but only \$10 if incapacitated. They make the decision to purchase insurance before the event occurs.

$$\rightarrow p = qb$$

(a) Show that with competitive insurance markets (and actuarially fair premiums) that any given individual will choose to fully insure.

$$E[V] = EV = (1-q) \cdot \log(w-p) + q \log(w-d-p+b)$$

$$EV = (1-q) \log(500 - qb) + q \log(10 - qb + b)$$

$= \$10 \Rightarrow w = \$500, d = \$490$

$$\frac{\partial EV}{\partial b} = (1-q) \cdot \frac{1}{500 - qb} \cdot (-q) + q \cdot \frac{1}{10 - qb + b} \cdot (-q + 1) = 0$$

$$\Rightarrow \frac{q(1-q)}{10 - qb + b} = \frac{q(1-q)}{500 - qb} \Rightarrow 500 - qb = 10 - qb + b$$

$$490 = b = d$$

Symmetry

- (b) What is the actuarially fair price for the insurance for each group? What is the expected utility of each group given that price if they buy insurance? What is the expected utility if they do not buy insurance? What prices are each groups willing to pay for the insurance? Assume insurers are perfectly competitive.

the actuarially fair price for the premium:

$$p = q \cdot b = q \cdot d = 490q$$

for low-risk types: $p = 490(0.02 + 0.03)$

$$p = 490(0.05) = \$24.5$$

for medium-risk types: $p = 490(0.02 + 0.08)$

$$p = 490(0.10) = \$49$$

for high-risk types: $p = 490(0.02 + 0.58)$

$$p = 490(0.60) = \$294$$

EU of each group if they buy insurance

low-risk: $EV = (1 - 0.05) \log(500 - 24.5) + 0.05 \log(500 - 490 - 24.5 + 490)$

$$EV = 0.95 \log(500 - 24.5) + 0.05 \log(500 - 24.5)$$

$$EV = \log(500 - 24.5) \rightarrow \log(W - \frac{p}{q})$$

$$EV = \log(475.5) \text{ [recall this is actually a natural log]}$$

$$EV = 6.16$$

fully insured

medium risk: $EV = \log(500 - 49) = \log(451) = 6.11$

high risk: $EV = \log(500 - 294) = \log(206) = 5.33$

EU of each group if they do not buy insurance

$$EV = (1 - q) \log(500) + q \log(10)$$

low-risk: $EV = 0.95 \log(500) + 0.05 \log(10) = 6.02$

medium-risk: $EV = 0.90 \log(500) + 0.10 \log(10) = 5.82$

high-risk: $EV = 0.4 \log(500) + 0.6 \log(10) = 3.87$

Prices each group is willing to pay for insurance

$$EV_{\text{insured}} \geq EV_{\text{uninsured}}$$

$$\log(500 - p) \geq (1 - q) \log(500) + q \log(10)$$

$$\log(500 - p) \geq \log(500^{1-q}) + \log(10^q)$$

$$\log(500 - p) \geq \log(500^{1-q} \cdot 10^q)$$

$$500 - p \geq 500^{1-q} \cdot 10^q$$

$$500 - 500^{1-q} \cdot 10^q \geq p$$

low risk

$$p \leq 500 - 500^{0.95} \cdot 10^{0.05}$$

$$p \leq 88.33$$

medium risk:

$$p \leq 500 - 500^{0.90} \cdot 10^{0.10}$$

$$p \leq 161.88$$

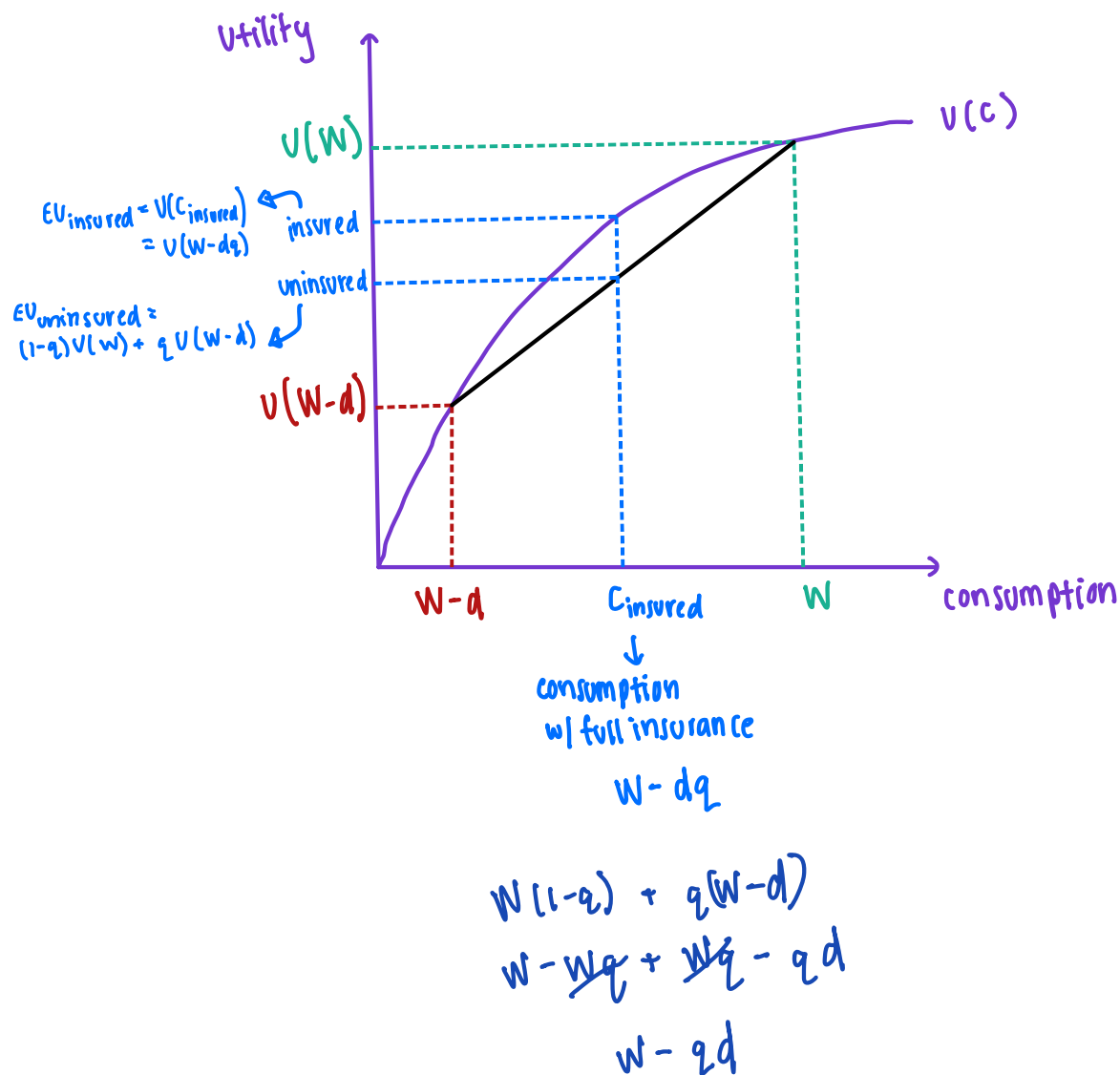
high risk:

$$p \leq 500 - 500^{0.4} \cdot 10^{0.6}$$

$$p \leq 452.18$$

(take everything to the $e^{\ln}$)

- (c) Show graphically how someone with a log utility (exhibiting diminishing marginal returns to consumption) can achieve a higher utility by insuring.



- (d) Now assume that the insurance company is not able to differentiate between the three types. Therefore, it offers a policy that fully insures individuals at the same price to all three types. Assuming all three types buy this insurance policy, what is the price for insurance?

assume

(1) firms make zero profits

(2) equal shares of each type in the population

$$\begin{aligned}
 p &= d \cdot E[\text{Pr}(\text{adverse event})] \\
 &= 490 \cdot \left[\frac{0.05 + 0.10 + 0.60}{3} \right] \\
 &= \$122.5
 \end{aligned}$$

- (e) Prove that with the price you found in part (d), the low-risk type would not be willing to buy the full insurance policy.

from (b) $EV_{\text{insured}} \geq EV_{\text{uninsured}}$

low risk

$$\begin{aligned}
 p &\leq 500 - 500 \cdot 10^{0.95} \cdot 0.05 \\
 p &\leq 88.33
 \end{aligned}$$

$\Rightarrow$ since in (d) we find $p = \$122.5$ which is higher than $\$88.33$, low-risk will not buy the new full insurance policy

- (f) Since the low-risk type does not buy the policy, the insurance company cannot offer the policy at the price from part (c). What is the new price for this policy if the risk averse drop out of the market? Who will buy at this price?

$$p = 490 \cdot \left[\frac{0.10 + 0.60}{2} \right] = \$171.75$$

who will buy @ this price?

will not buy

medium risk:

$$\begin{aligned}
 p &\leq 500 - 500^{0.90} \cdot 10^{0.10} \\
 p &\leq 161.88
 \end{aligned}$$

will buy

high risk:

$$\begin{aligned}
 p &\leq 500 - 500^{0.4} \cdot 10^{0.6} \\
 p &\leq 452.18
 \end{aligned}$$

the market has unraveled!

however since medium risk will not buy the $p = 490 \cdot 0.60 = \$294$ (the actuarially price)

(g) You have demonstrated an example of a market unraveling. Explain the intuition for why it happened.

(h) How would results change if the individuals' income is zero when incapacitated?